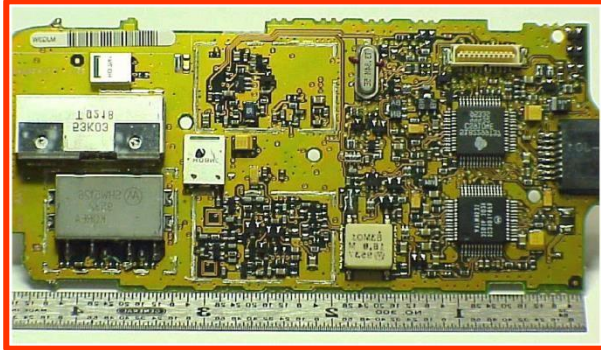




Antennas: Principle and Design

Introduction

Two-way wireless transmission



Cellular phone handset

Key component: Antenna

Antenna:

as an interface

as an impedance transformer



Base station Tower

Introduction



IEEE Definition of an antenna

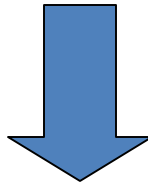
- Antenna is that part of a transmitting or receiving system which is designed to radiate or to receive EM waves.

Proper design of this part is necessary for efficient coupling

How do Antennas form and Radiate Electromagnetic Fields



The accelerating charge radiate energy i.e they throw energy in their surrounding medium



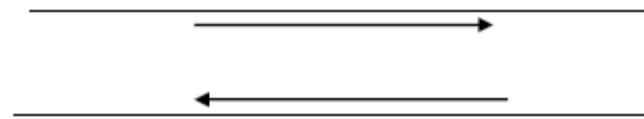
RADIATION

Radiation is a phenomenon related to time-varying currents.

How do Antennas form and Radiate Electromagnetic Fields



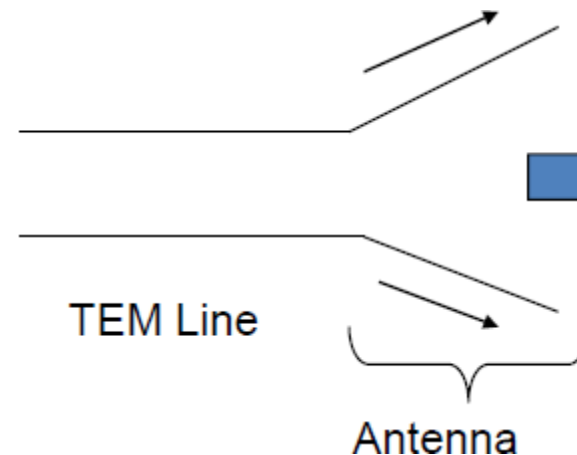
Case I: $d \ll \lambda$



No net radiation.

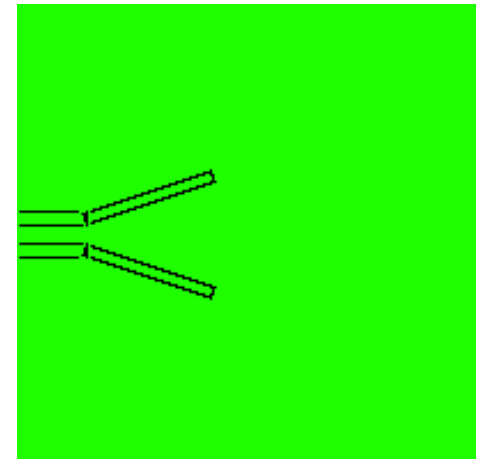
[Note: In electromagnetics: (physical closeness) measured in terms of λ . If $d \ll \lambda$ radiation is negligible]

Case II: $d \approx \lambda$

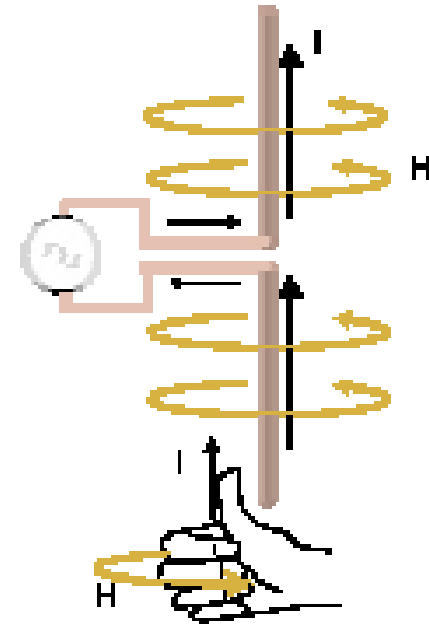
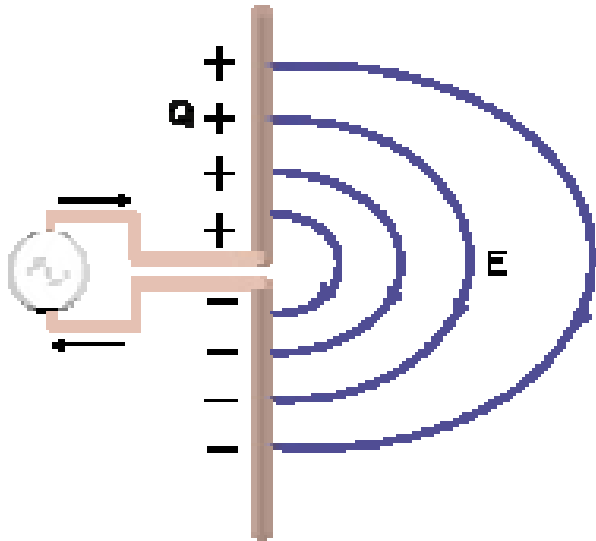


Radiation

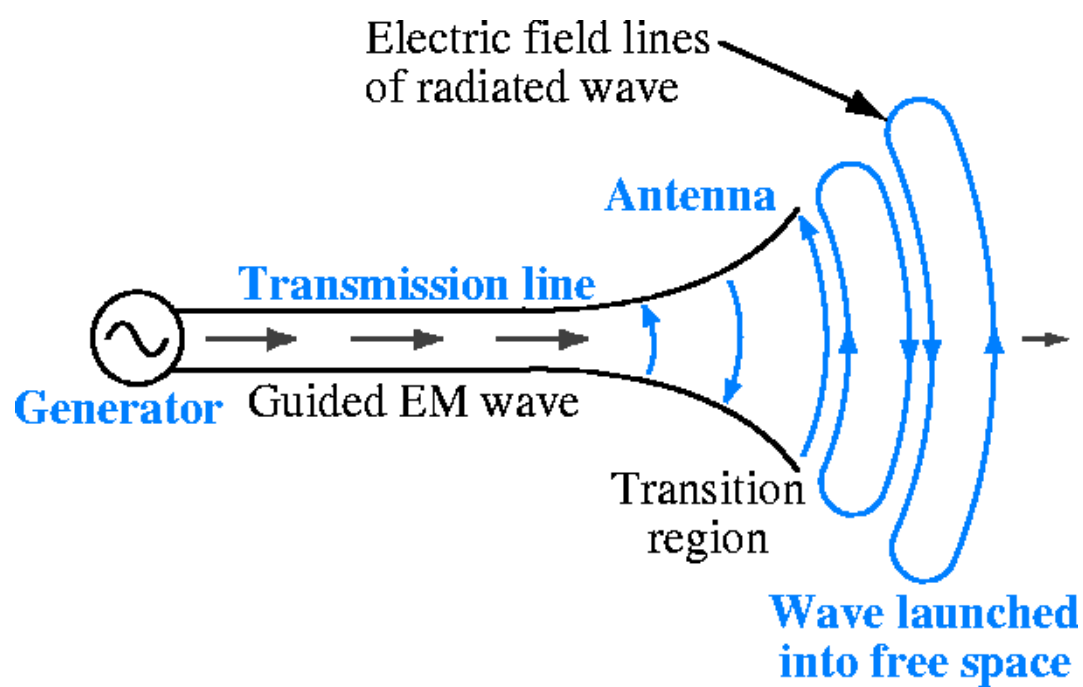
$d \approx \lambda$



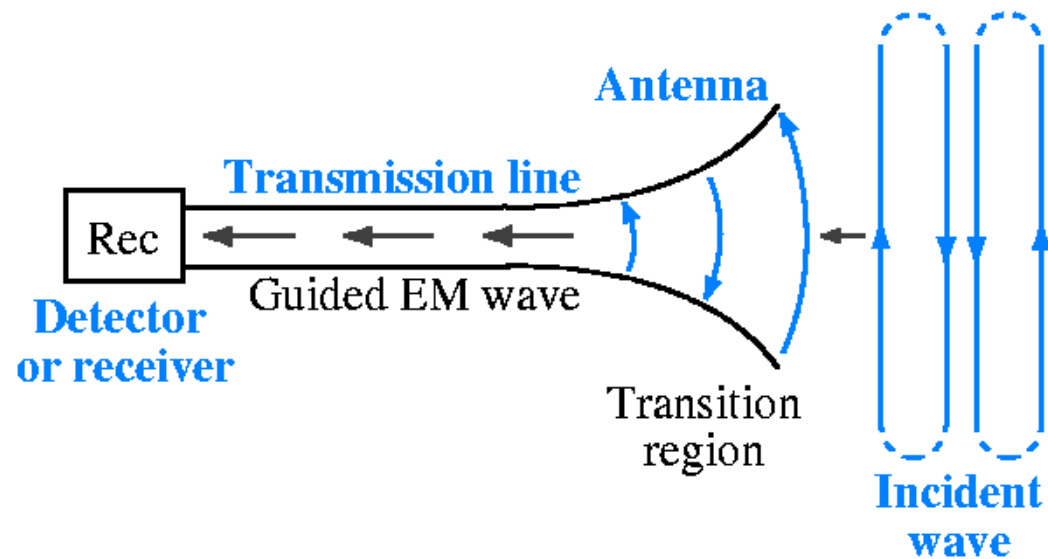
How do Antennas form and Radiate Electromagnetic Fields



The charge and current on the dipole create fields that are perpendicular to each other.



(a) Transmission mode

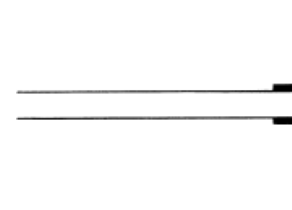


(b) Reception mode

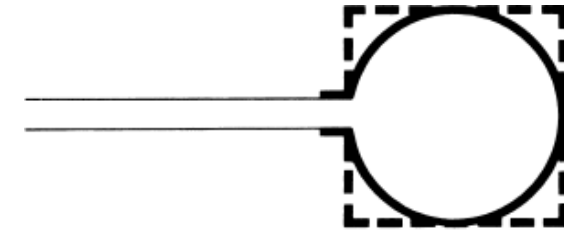
Types of Antennas



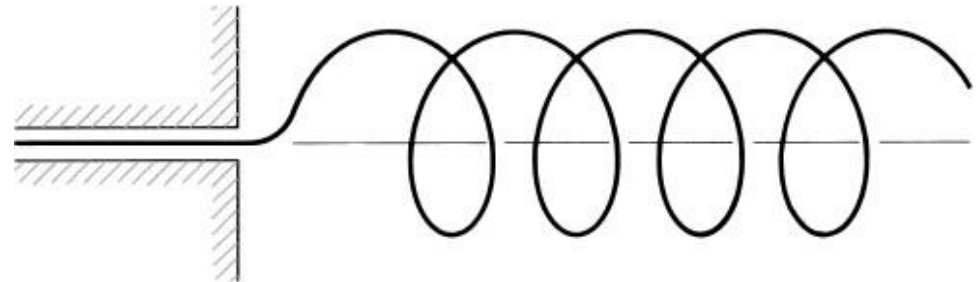
1. Wire Antennas
2. Aperture Antennas
3. Microstrip Antennas



(a) Dipole



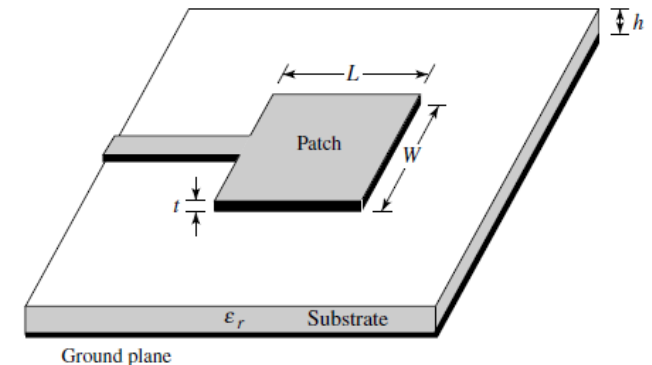
(b) Circular (square) loop



Pyramidal horn antenna



Conical horn antenna



Types of Antennas



- 4. Array Antennas
- 5. Broadband Antennas
- 6. Frequency Independent Antennas

An ideal antenna is one that will radiate all the power delivered to it from transmitter in a desired directions.

Fundamental Parameters of Antennas



To describe the performance of an antenna, definitions of various parameters are necessary.

An antenna *radiation pattern* or *antenna pattern* is defined as “a mathematical function or a graphical representation of the radiation properties of the antenna as a function of space coordinates”.



Mostly in Far-field Region



To Study Radiation properties we need convenient set of coordinates

Fundamental Parameters of Antennas

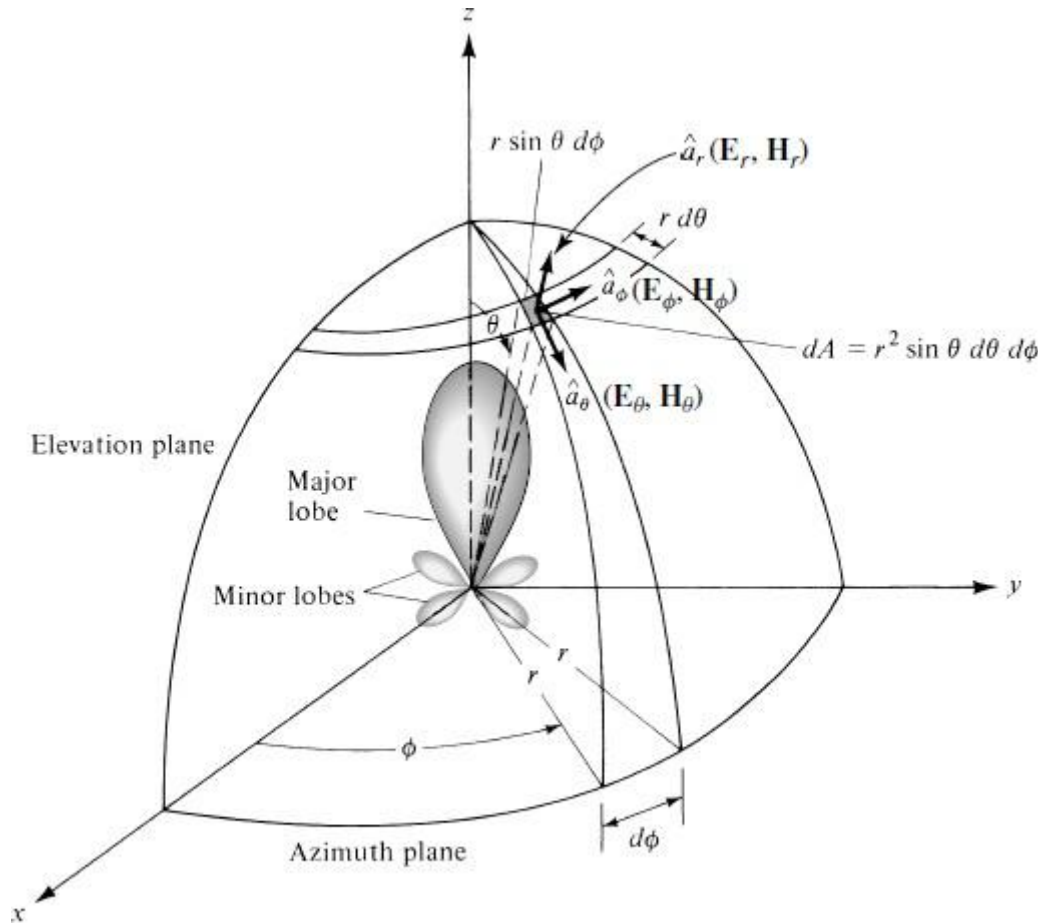


Figure 2.1 Coordinate system for antenna analysis.

Radiation Pattern are of two types;

- Field pattern
- Power pattern

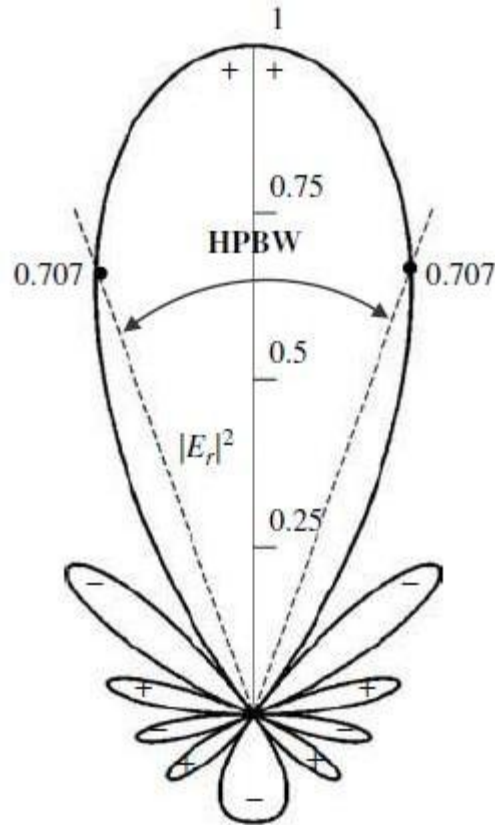
Radiation Pattern Includes:

- Major Lobes
- Minor Lobes
- Side Lobes
- Back Lobes

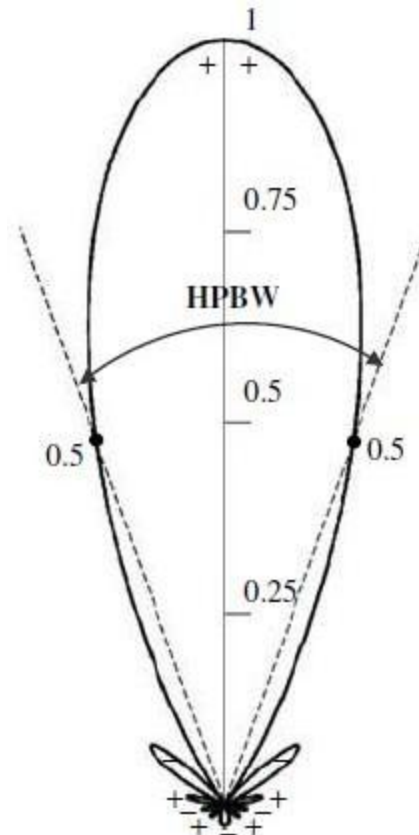
$$E_{\theta}(\theta, \phi)_n = \frac{E_{\theta}(\theta, \phi)}{E_{\theta}(\theta, \phi)_{\max}}$$

$$P_n(\theta, \phi) = \frac{P(\theta, \phi)}{P(\theta, \phi)_{\max}}$$

2D-Radiation patterns

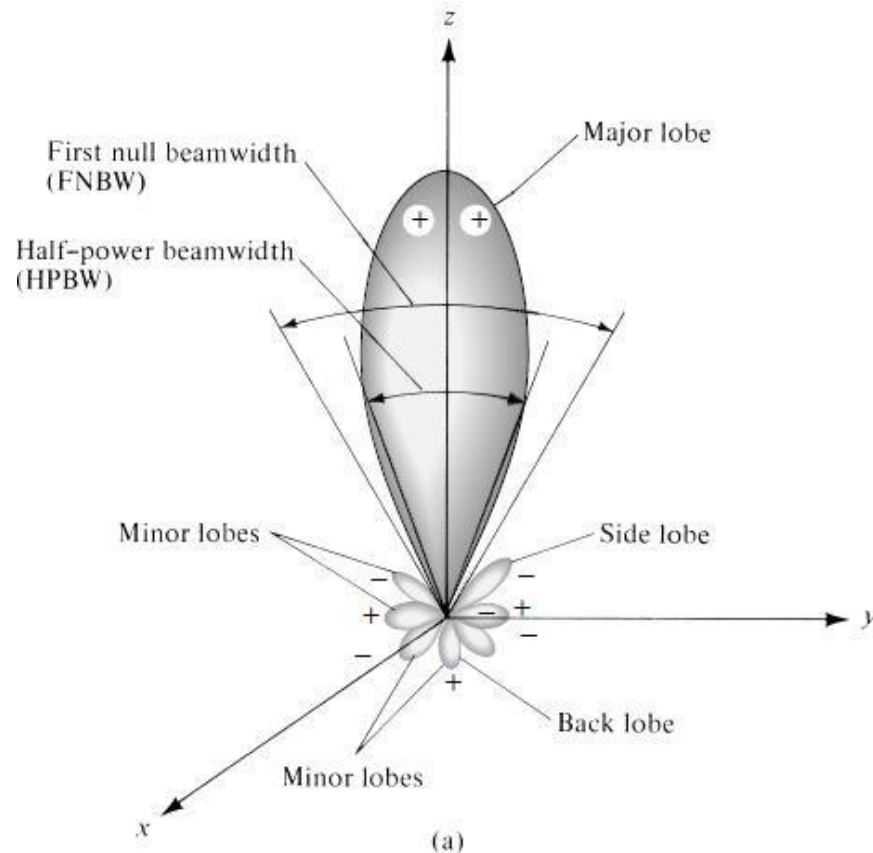


(a) Field pattern (in linear scale)



(b) Power pattern (in linear scale)

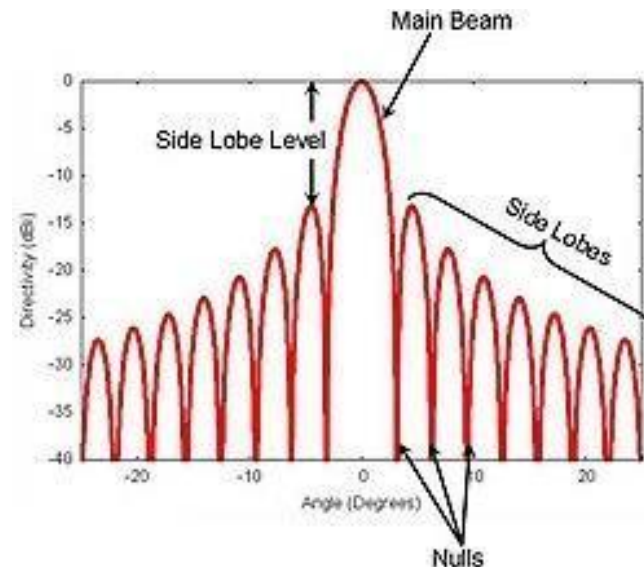
Radiation pattern Lobes



- Major Lobe (Main Beam): contains direction of maximum radiation (maybe more than one as in split beam antenna).
- Minor Lobe: any lobe except a major lobe.
- Side Lobe: other than intended lobe.
- Back Lobe: 180° angle with respect to antenna beam.

Minor lobes are undesired. Side lobes are the largest lobes of minor lobes. *Side Lobe Level, SLL*

The ratio, usually expressed in decibels (dB), of the amplitude at the peak of the main lobe to the amplitude at the peak of a side lobe.

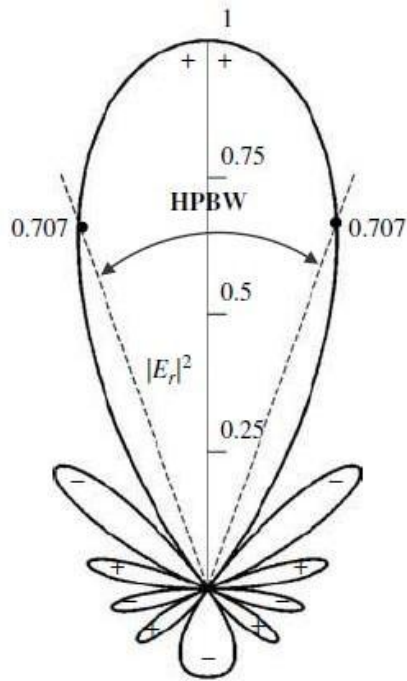


In most radar systems, low side lobe ratios are very important to minimize false target indications through the side lobes.

Normalized Pattern

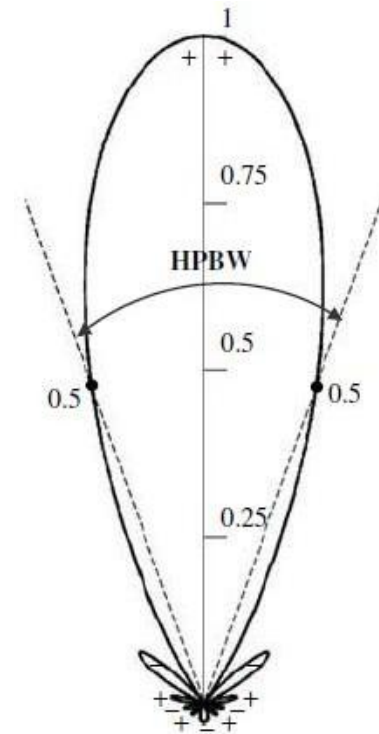


Usually, the pattern describes the normalized field (power) values with respect to the maximum value.



(a) Field pattern (in linear scale)

$$E_{\theta}(\theta, \phi)_n = \frac{E_{\theta}(\theta, \phi)}{E_{\theta}(\theta, \phi)_{\max}}$$
$$P_n(\theta, \phi) = \frac{P(\theta, \phi)}{P(\theta, \phi)_{\max}}$$

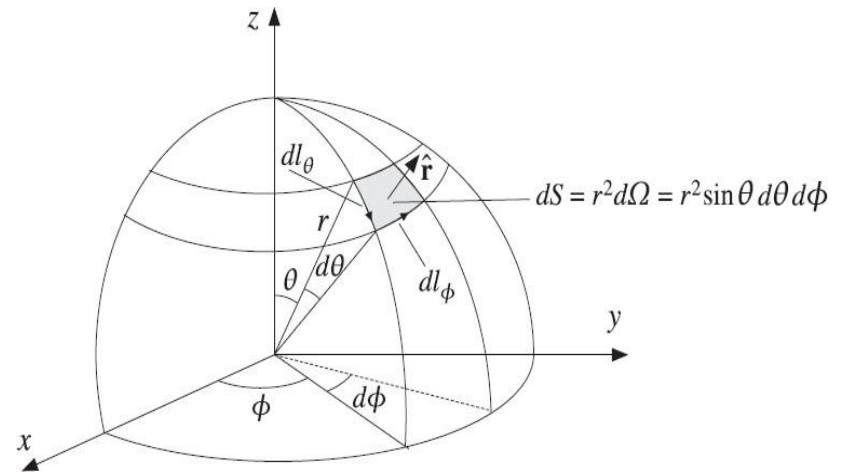
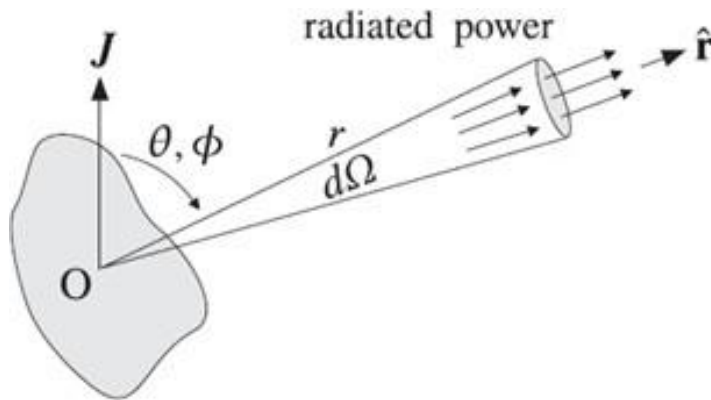


(b) Power pattern (in linear scale)

Antenna Parameters



Radiation Intensity is defined as the power radiated per unit solid angle
The **total radiated power** (P_{rad}) is obtained by integrating $P(\theta, \phi)$ over all solid angle



Useful concept: $\rightarrow$ Isotropic radiator $\rightarrow$ **Isotropic radiation intensity** P_i .

P_i is the average of the radiation intensity over all solid angles.

Isotropic, and Directional Patterns

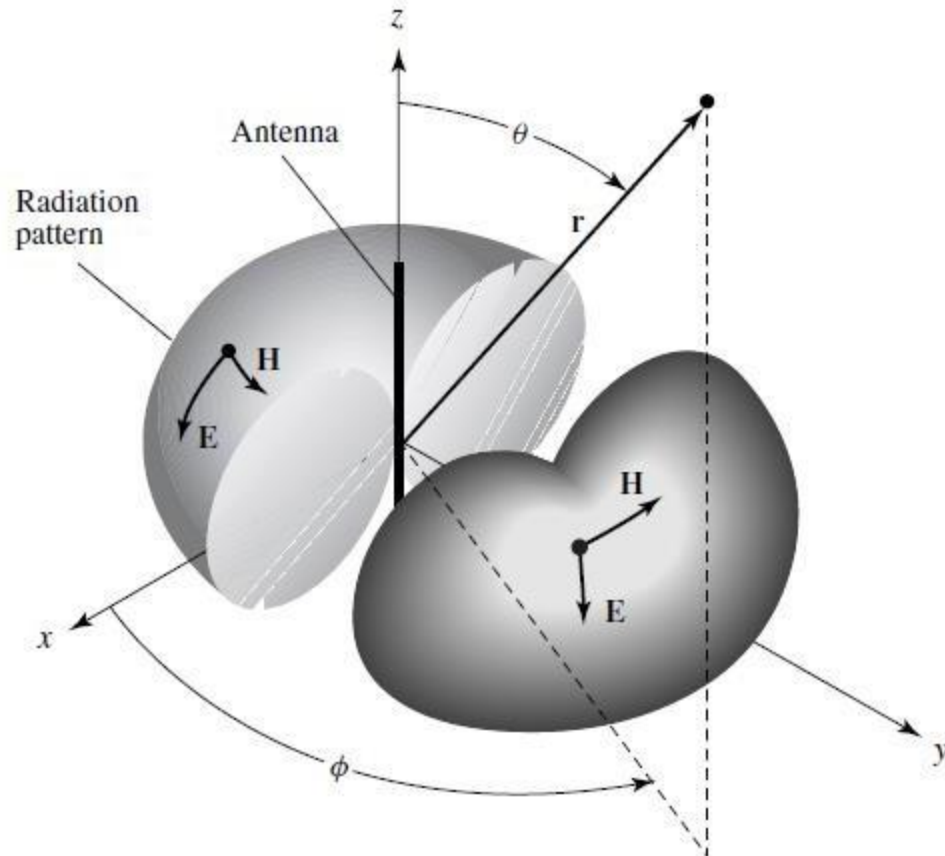


Isotropic radiator:

A hypothetical, lossless antenna having equal radiation in all direction. Not realized but used as a reference.

Directional antenna:

Radiating or receiving EM waves more effectively in some directions than in others.



Hertzian Dipole



Retarded Potential $\vec{A}$

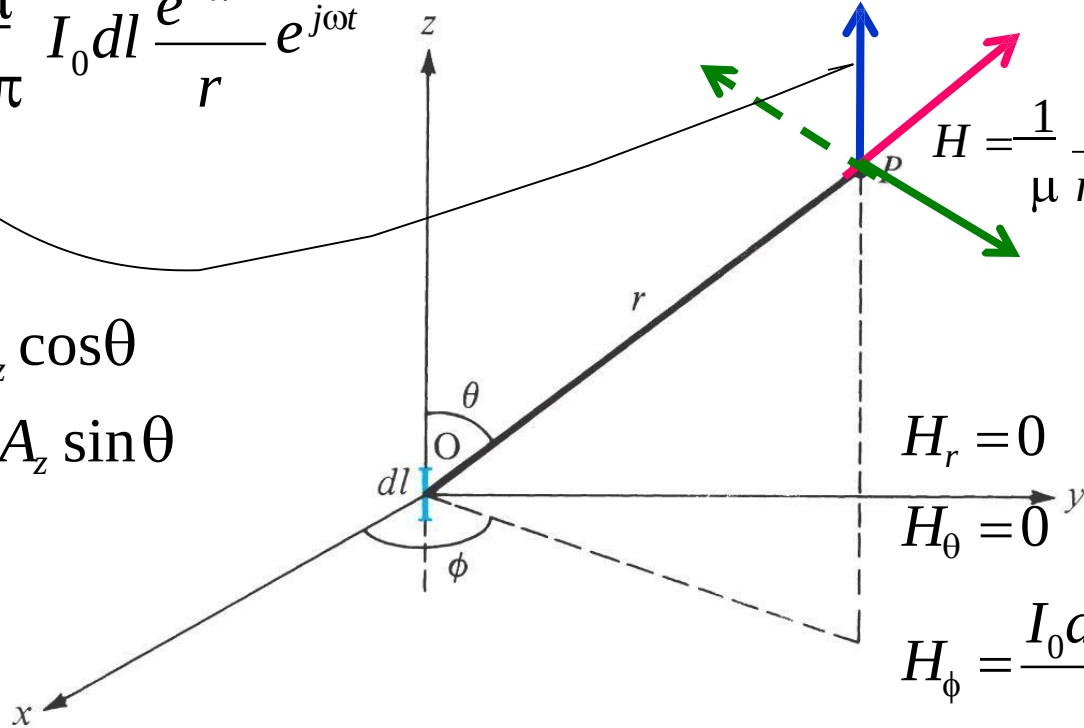
$$\vec{H} = \frac{1}{\mu} \nabla \times \vec{A}$$

$$\vec{A}_z = \frac{\mu}{4\pi} I_0 dl \frac{e^{-j\beta r}}{r} e^{j\omega t}$$

$$A_{rs} = A_z \cos\theta$$

$$A_{\theta s} = -A_z \sin\theta$$

$$A_{\phi s} = 0$$



$$H = \frac{1}{\mu} \frac{1}{r^2 \sin\theta}$$

$$\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \partial/\partial r & \partial/\partial\theta & 0 \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

$$H_r = 0$$

$$H_\theta = 0$$

$$H_\phi = \frac{I_0 dl \sin\theta}{4\pi} e^{j(\omega t - \beta r)} \left[\frac{j\beta}{r} + \frac{1}{r^2} \right]$$



Hertzian Dipole



$$H_r = 0$$

$$H_\theta = 0$$

$$H_\phi = \frac{I_0 dl \sin \theta}{4\pi} e^{j(\omega t - \beta r)} \left[\frac{j\beta}{r} + \frac{1}{r^2} \right]$$

1/r³ : electrostatic field

1/r²: induction field

1/r: radiation field

$$\vec{E} = \frac{1}{j\omega\epsilon} \nabla \times \vec{H} = \frac{1}{j\omega\epsilon} \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r \sin \theta \hat{a}_\phi \\ \partial/\partial r & \partial/\partial \theta & 0 \\ H_r & rH_\theta & r \sin \theta H_\phi \end{vmatrix}$$

$\beta r \ll 1$: near field zone condition

$\beta r \gg 1$: far field zone condition

$$E_r = \frac{\eta I_0 dl \cos \theta}{2\pi} e^{j(\omega t - \beta r)} \left[\frac{1}{r^2} - \frac{j}{\beta r^3} \right]$$

$$E_\theta = \frac{\eta I_0 dl \sin \theta}{4\pi} e^{j(\omega t - \beta r)} \left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3} \right]$$

where,

$$\eta = \frac{\beta}{\omega\epsilon}$$

Hertzian Dipole

$$P_{rad} = \frac{1}{2} \int \text{Re}(\vec{E}_{\theta s} \times \vec{H}_{\phi s}^*) \cdot d\vec{S} = \int \frac{1}{2} \eta |H_{\phi s}|^2 \cdot dS$$

$$H_{\phi s} = \frac{I_0 dl \sin \theta}{4\pi} e^{-j\beta r} \frac{j\beta}{r}$$

$$P_{rad} = 40\pi^2 \left[\frac{dl}{\lambda} \right]^2 I_0^2$$

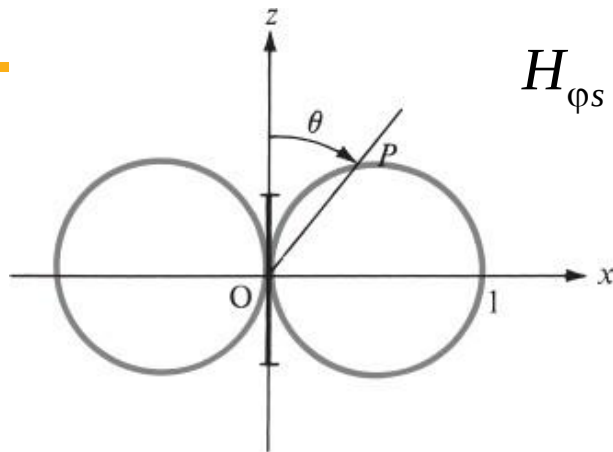
$$R_{rad} = \frac{2P_{rad}}{I_0^2}$$

Radiation resistance

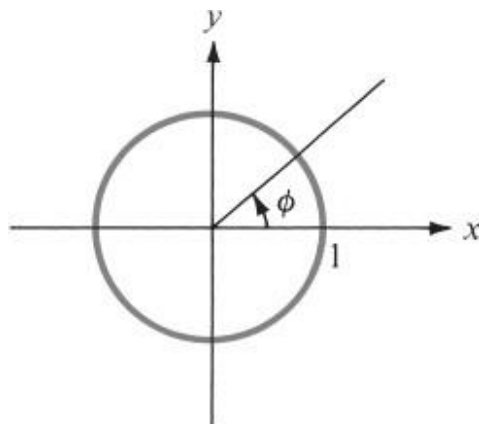
$$R_{rad} = 80\pi^2 \left[\frac{dl}{\lambda} \right]^2$$



Field Pattern of Hertzian dipole



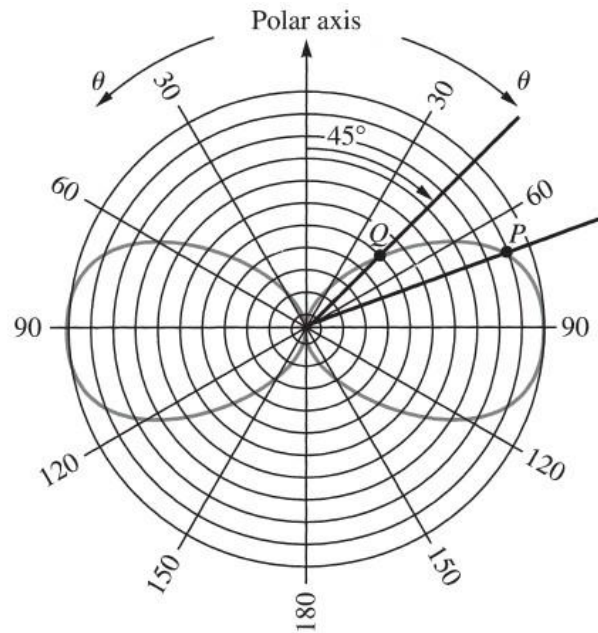
(a)



(b)

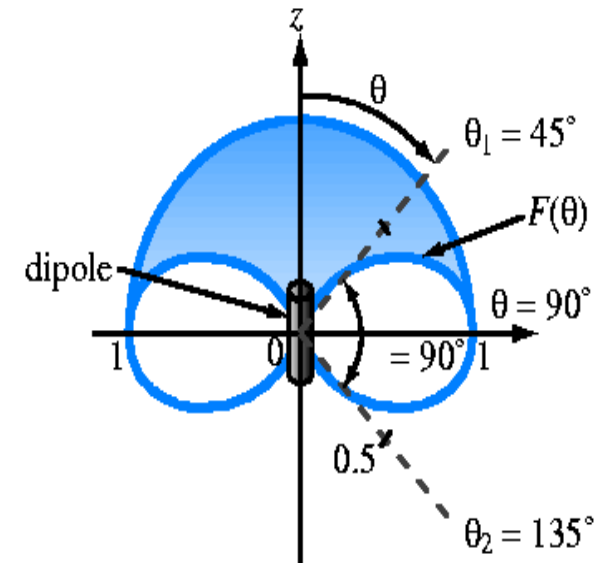
$$H_{\phi s} = \frac{I_0 dl \sin \theta}{4\pi} e^{-j\beta r} \frac{j\beta}{r} \quad E_{\theta s} = \frac{\eta I_0 dl \sin \theta}{4\pi} e^{-j\beta r} \frac{j\beta}{r}$$

- Directivity
- HPBW
- Effective area (or aperture)

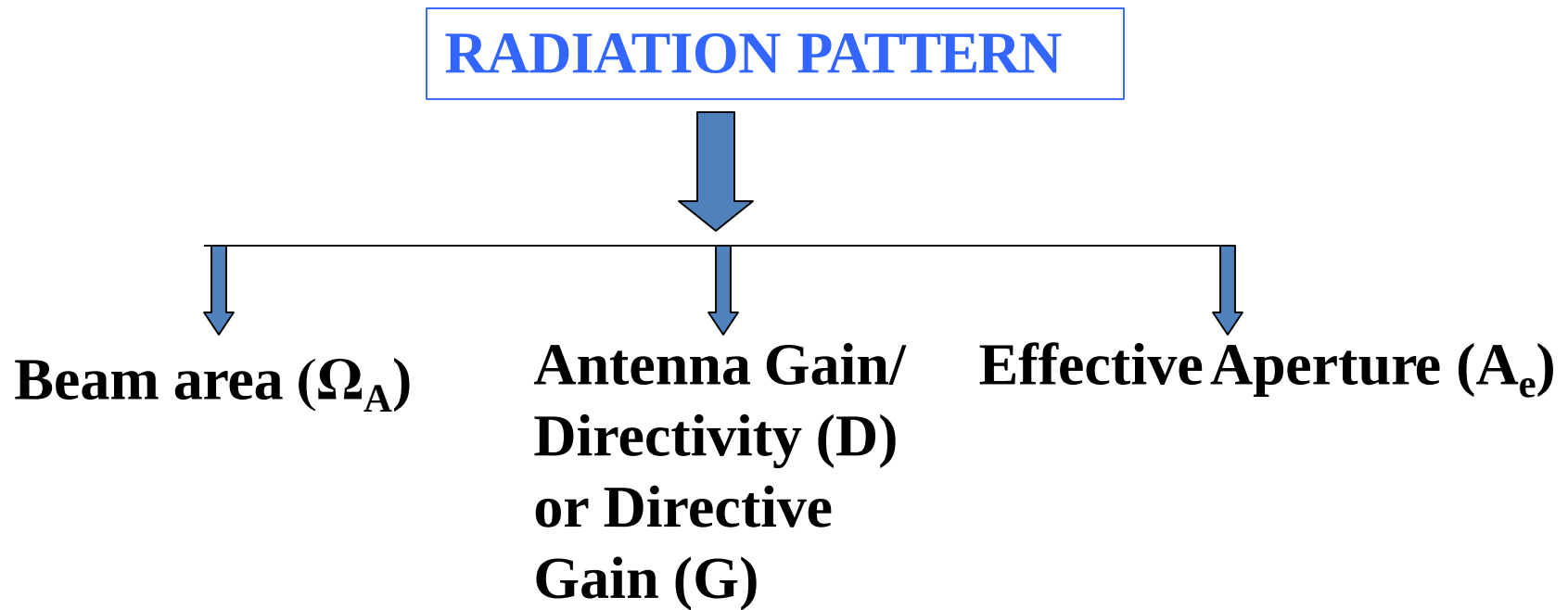


(a)

Power Pattern



Other Radiation Properties



Radiation Properties



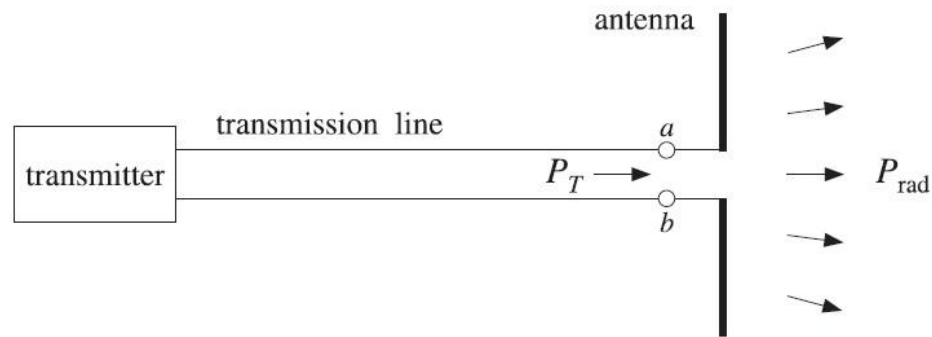
Directive gain: Radiation intensity normalized by the corresponding isotropic intensity.

- It measures the ability of the antenna to direct its power towards a given direction.

Directivity: The maximum value of the directive gain.

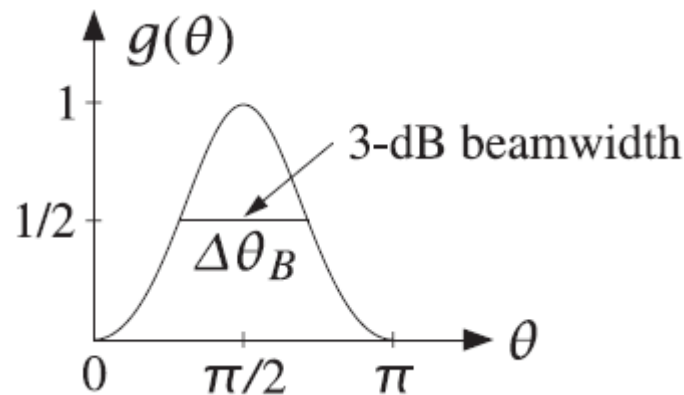
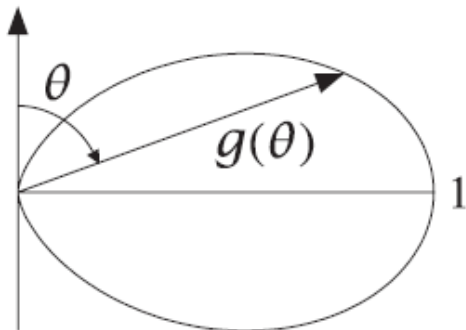
Power gain (or Gain of an antenna): Radiation intensity normalized by the total power P_T accepted by the antenna terminals.

- Antenna efficiency



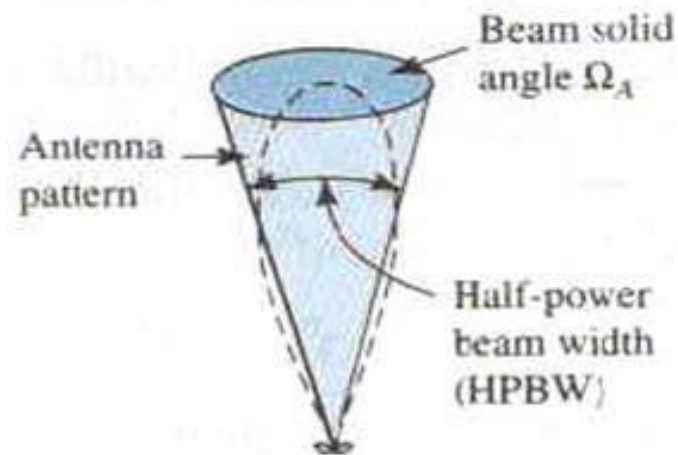
$$g(\theta, \phi) = \frac{G(\theta, \phi)}{G_{max}}$$

(normalized gain)

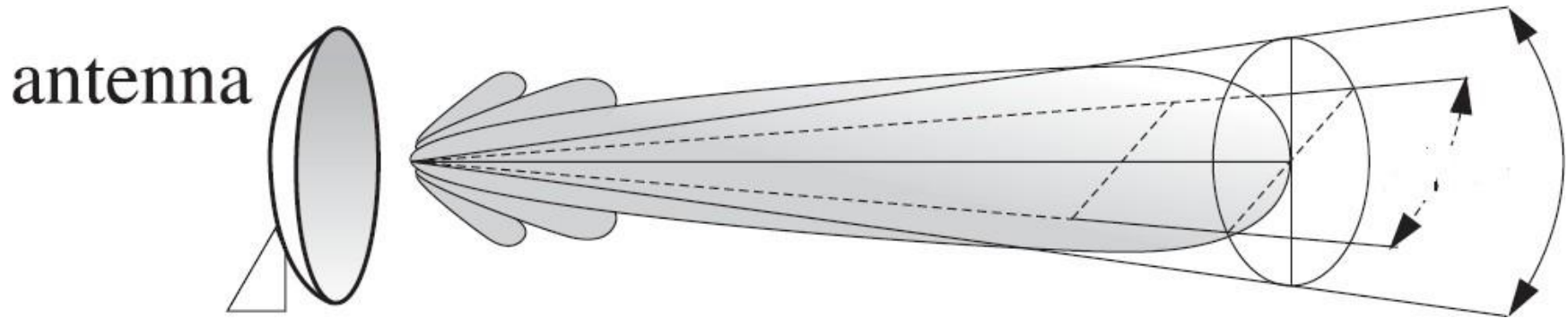


Beam Area

$$\Omega_A = \int_0^\pi \int_0^{2\pi} g(\theta, \phi) d\Omega = \int_0^\pi \int_0^{2\pi} \text{Normalised power pattern}$$



Concept of Approximate Directivity



$$D = \frac{4\pi}{\Omega_A}$$

Exact Directivity

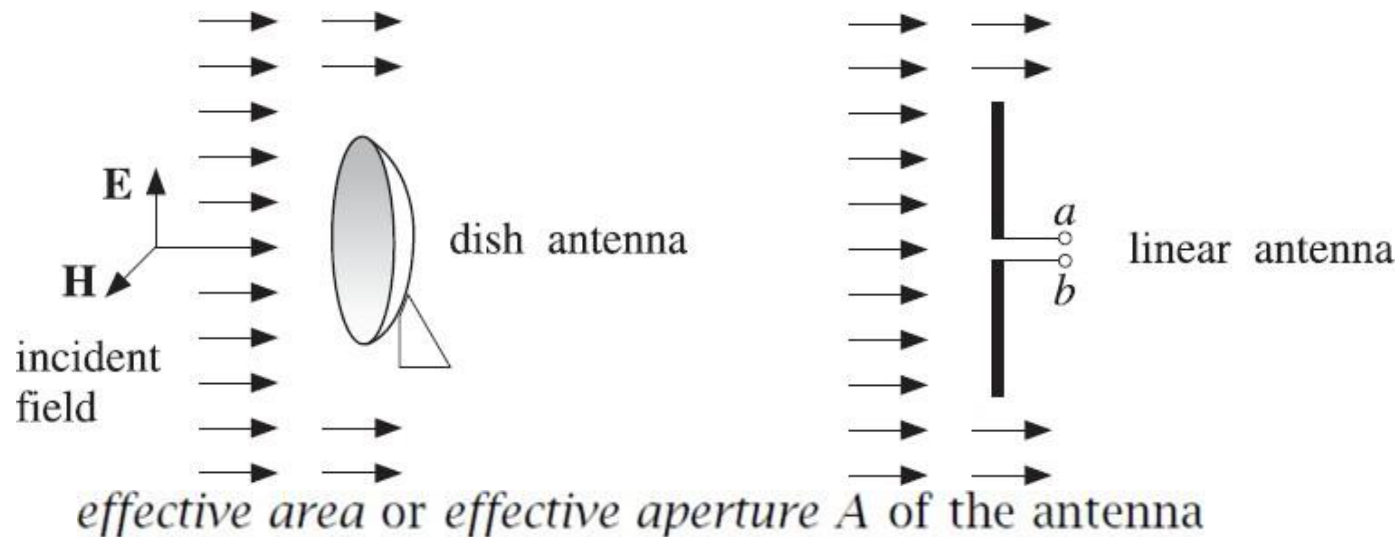
$$\Omega_A = 4\pi$$

$$D = 1$$

For an isotropic antenna

Effective Aperture (or Effective area)

When an antenna is operating as a receiving antenna, it extracts a certain amount of power from an incident electromagnetic wave.



power P_R available at the antenna output terminals $P_R = A \mathcal{P}_{\text{inc}}$

incident power density $\mathcal{P}_{\text{inc}} = E^2 / 2\eta$

$$G(\theta, \phi) = \frac{4\pi A(\theta, \phi)}{\lambda^2}$$

Power Gain:

$$G_p(\theta, \phi) = \frac{4\pi A(\theta, \phi)}{\lambda^2}$$

The effective area is not equal to the physical area of an antenna

$$A = e_a \frac{1}{4} \pi d^2 \quad (\text{effective area of dish antenna})$$

where e_a is the *aperture efficiency* factor, typically $e_a = 0.55\text{--}0.65$.

Determine the gain in dB of a dish antenna of diameter of 0.5 m operating at a satellite downlink frequency of 4 GHz and having 60% aperture efficiency. Repeat if the downlink frequency is 11 GHz. Repeat if the diameter is doubled to 1 m.

	$d = 0.5 \text{ m}$	$d = 1 \text{ m}$
$f = 4 \text{ GHz}$	263 = 24 dB	1052 = 30 dB
$f = 11 \text{ GHz}$	1990 = 33 dB	7960 = 39 dB

Bandwidth



- The bandwidth of an antenna is defined as “the range of frequencies within which the performance of the antenna, with respect to some characteristic, conforms to a specified standard.”
- **For broadband antennas:** the bandwidth is usually expressed as the ratio of the upper-to-lower frequencies of acceptable operation.
- **For narrowband antennas:** the bandwidth is expressed as a percentage of the frequency difference (upper minus lower) over the center frequency of the bandwidth.

Important Characteristics of Antennas:

- ➡ Radiated Power (P_T)
- ➡ Received Power (P_R)
- ➡ Gain ($G_{R,T}$) (relative to reference)
- ➡ Wavelength (λ)
- ➡ Polarization (vertical vs. horizontal)
- ➡ Distance from Source (R)

Many of these terms come together in the **Friis** Equation.

The Friis Transmission Formula:

$$P_R = [P_T G_T G_R \lambda^2] / [4\pi R]^2$$

- ➡ From this it can be seen that received antenna power increases with
 1. Transmitted power
 2. Gains (directionality relative to each other)
 3. Wavelength
- ➡ And it decreases with distance².

Decibel Power Units:



➡ Antenna power levels vary by *many* orders of magnitude. To deal with that we use units of **decibels** to describe power and relative power (gain).

In equation form a decibel is given by:

$$P_{dB} = 10 \log_{10}(P)$$

➡ An example:

- 1) A 100 kW transmitter produces 10^5 Watts of output
- 2) $\log_{10}(10^5) = 5$
- 3) Therefore $P_{dB} = 10 \times 5 = 50 \text{ dB!}$

How to determine the radiation pattern?

Vector potential approach

Antenna Analysis:

- (a) Determination of current distribution on the antenna structure.
- (b) Determine $\mathbf{A}$ from $\mathbf{J}$
Field due to this current distribution in the space surrounding the antenna
 - Find $\mathbf{H}$ from $\mathbf{A}$
 - Find $\mathbf{E}$ from $\mathbf{H}$
- (f) Calculate the power density $\mathbf{P}$

An Overview of Antennas:



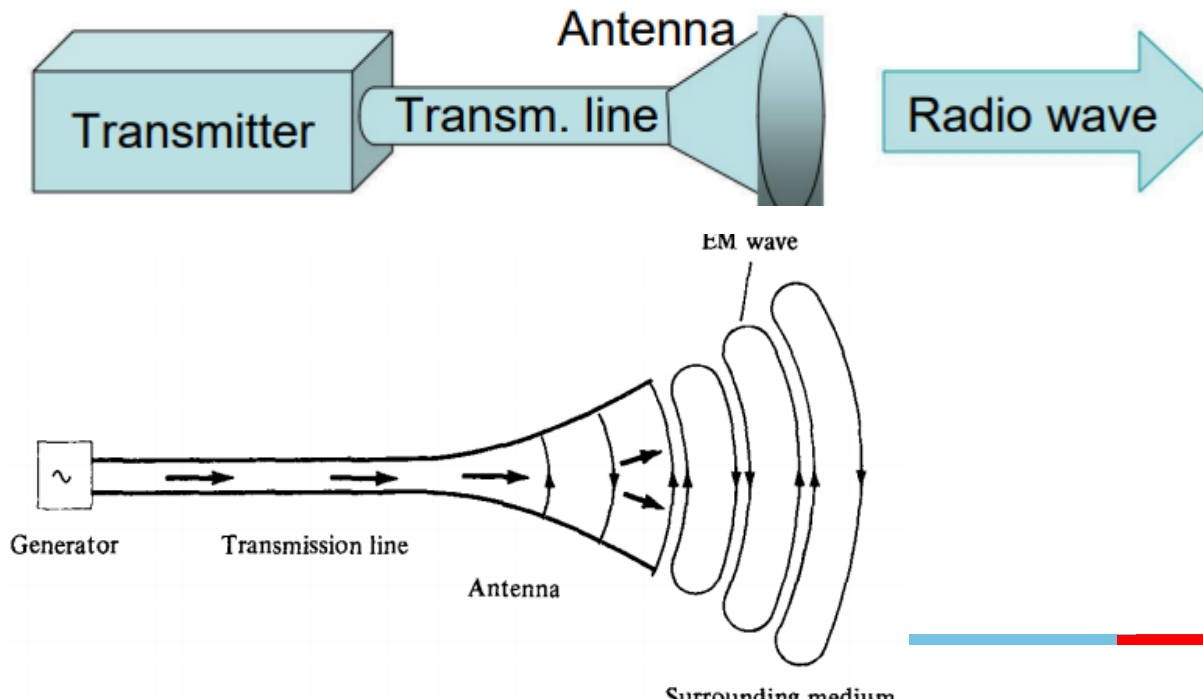
WHY ANTENNAS ?

- Need of antenna arisen when two person wanted to communicate between them when separated by some distance and wired communication is not possible.
- Antennas are required by any radio receiver or transmitter to couple its electrical connection to the electromagnetic field.
- Radio waves are electromagnetic waves which carry signals through the air (or through space) at the speed of light with almost no transmission loss.
- In each and every case, the transmitters and receivers involved require antennas, although these are sometimes hidden (such as the antenna inside an AM radio or inside a laptop computer equipped with wi-fi).

What is an Antenna?



- ➡ An antenna is a device that radiates or receives RF signals.
- ➡ An antenna can be as simple as a single wire or as complex as a large dish. It depends on the application!

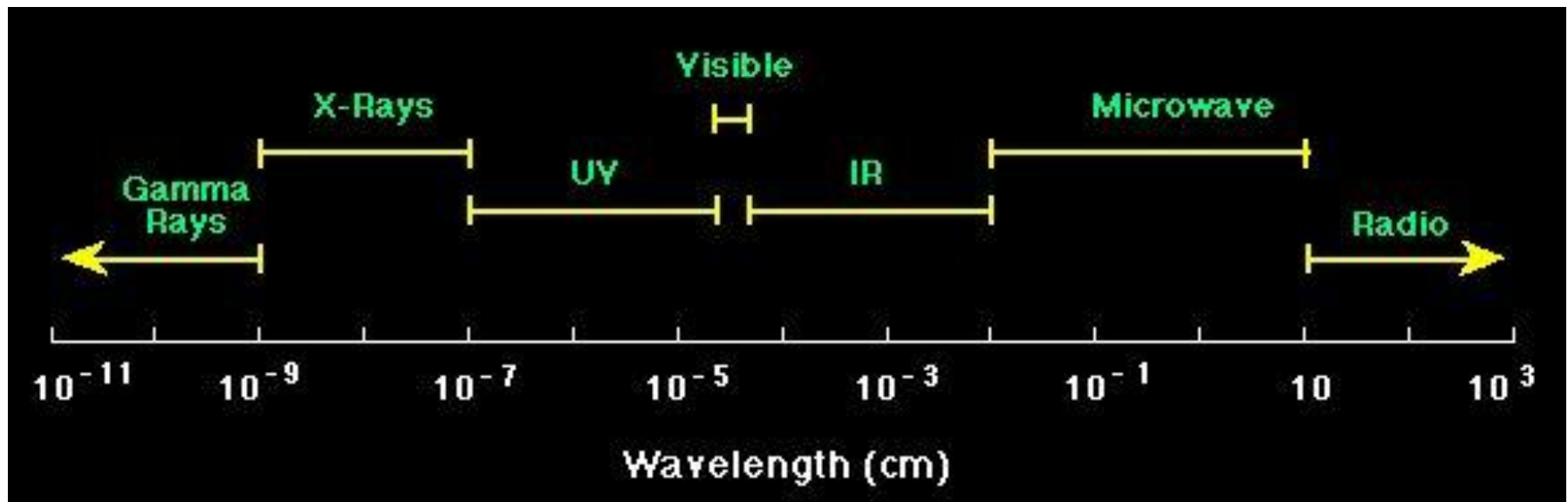


The Antenna Reciprocity Principle:



- ➡ An antenna behaves the same whether it is radiating or receiving RF signals.
- ➡ The difference depends on HOW it's used and in WHAT is put at the back end (electronics).
- ➡ The basic principle of a **transmitting** antenna is that an oscillating current in the device creates an oscillating electro-magnetic field.
- ➡ The basic principle of a **receiving** antenna is that an oscillating electro-magnetic field will trigger an oscillating current.

What are Radio Frequencies (RF)?



How long is long?



Wavelength x freq. = c

where $c = 3 \times 10^{10}$ cm/sec

➡ A typical radio station (AM) broadcasts at 1 MHz (10^6).
This gives a wavelength ~ **300 meters!**

Frequencies and Wavelengths



VLF	LF	MF	HF	VHF	UHF	SHF	EHF
3	30	300	3	30	300	3	30
kHz	kHz	kHz	MHz	MHz	MHz	GHz	GHz
100	10	1	100	10	1	10	1
km	km	km	m	m	m	cm	cm

➡ The type of antenna used is very dependent on the bandwidth sampled and the wavelength of the RF.

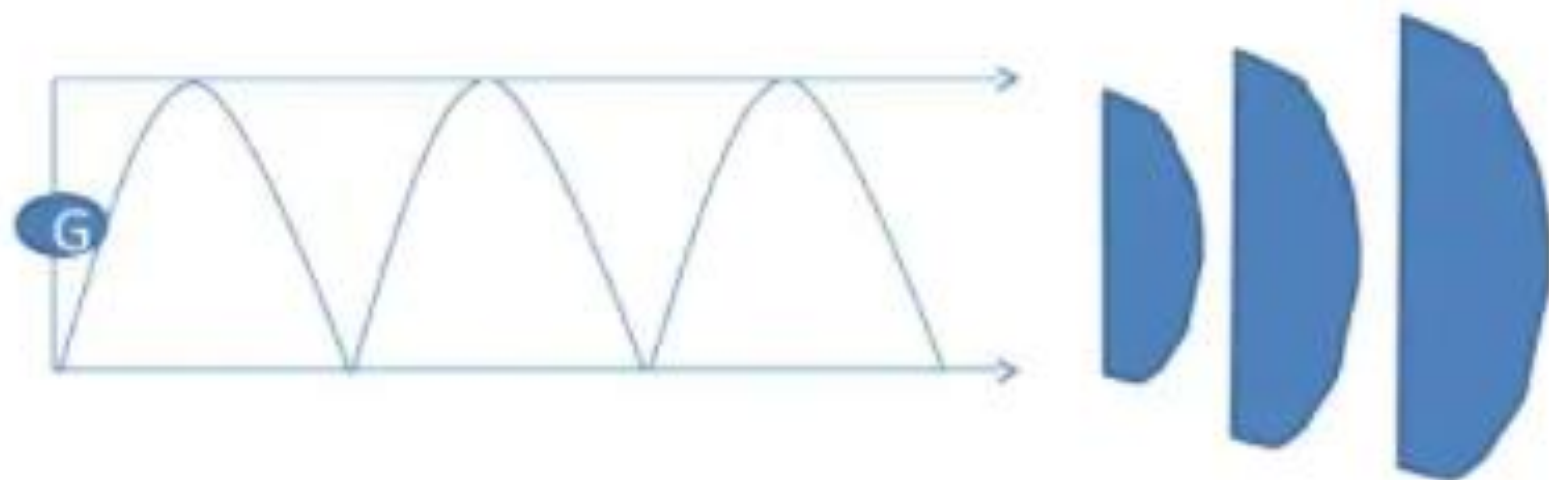
Some common RF bands.



- 1) AM Radio: 0.50 $\Rightarrow$ 1.7 MHz
- 2) SW Radio: 5.90 $\Rightarrow$ 29.1 MHz
- 3) CB Radio: 26.9 $\Rightarrow$ 27.4 MHz
- 4) FM Radio: 88.0 $\Rightarrow$ 108.0 MHz
- 5) TV VHF (low): 54.0 $\Rightarrow$ 88.0 MHz
- 6) TV VHF (high): 174 $\Rightarrow$ 220 MHz
- 7) Wireless Phone: 900 MHz
- 8) GPS: 1.25 $\Rightarrow$ 2.58 GHz
- 9) Deep Space Network: 2.29 $\Rightarrow$ 2.31 GHz
- 10) WiFi: 2.4 GHz

RADIATION MECHANISM

- Ideally all incident energy must be reflected back when open circuit. But practically a small portion of electromagnetic energy escapes from the system that is it gets radiated.
- This occurs because the line of force don't undergo complete phase reversal and some of them escapes.



RADIATION MECHANISM

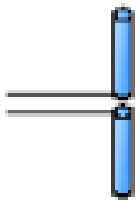
achieve

lead

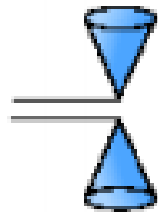
- To increase amount of radiated power open circuit must be enlarged , by spreading the two wires.
- Due to this arrangement, coupling between transmission line and free space is improved.
- Also amount of cancellation has reduced.
- The radiation efficiency will increase further if two conductors of transmission line are bent so as to bring them in same line.



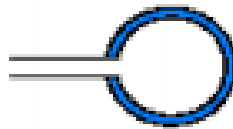
Different Antennas



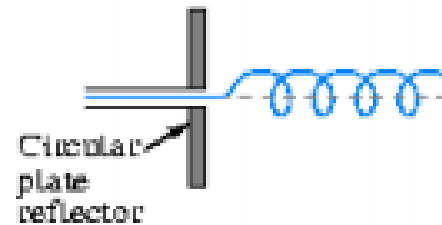
(a) Thin dipole



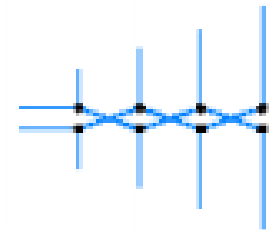
(b) Biconical dipole



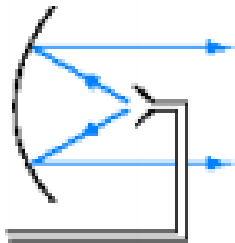
(c) Loop



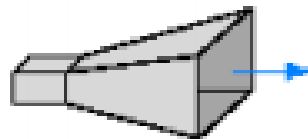
(d) Helix



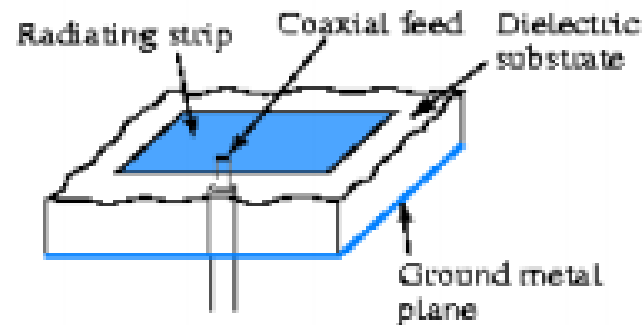
(e) Log-periodic



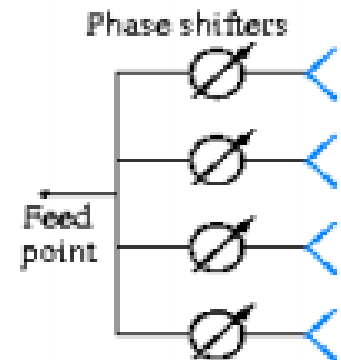
(f) Parabolic dish reflector



(g) Horn



(h) Microstrip



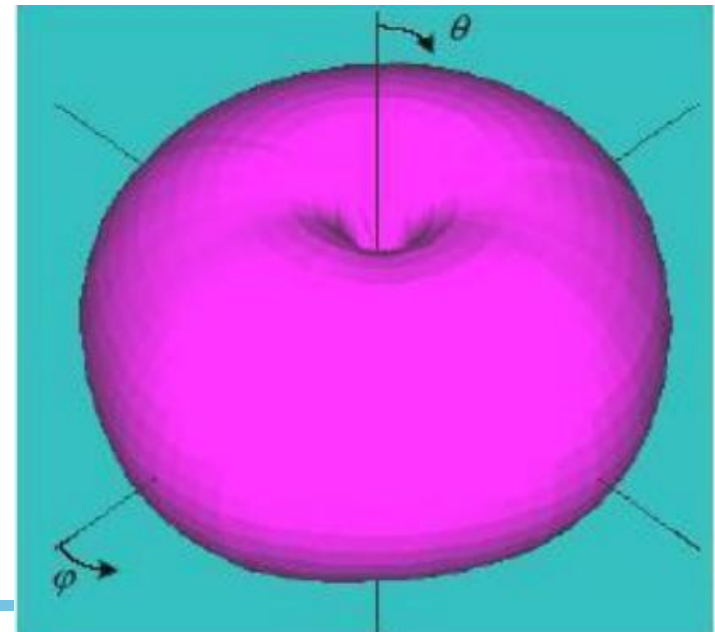
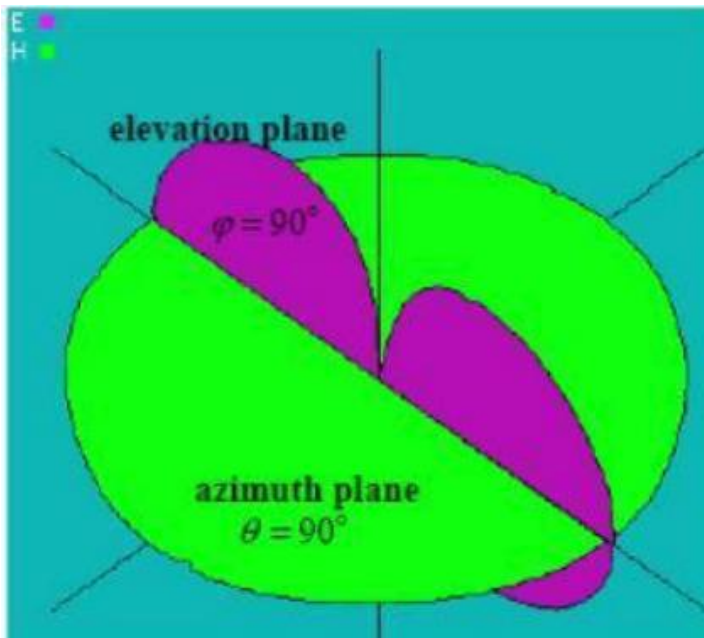
(i) Antenna array

TYPES OF ANTENNAS

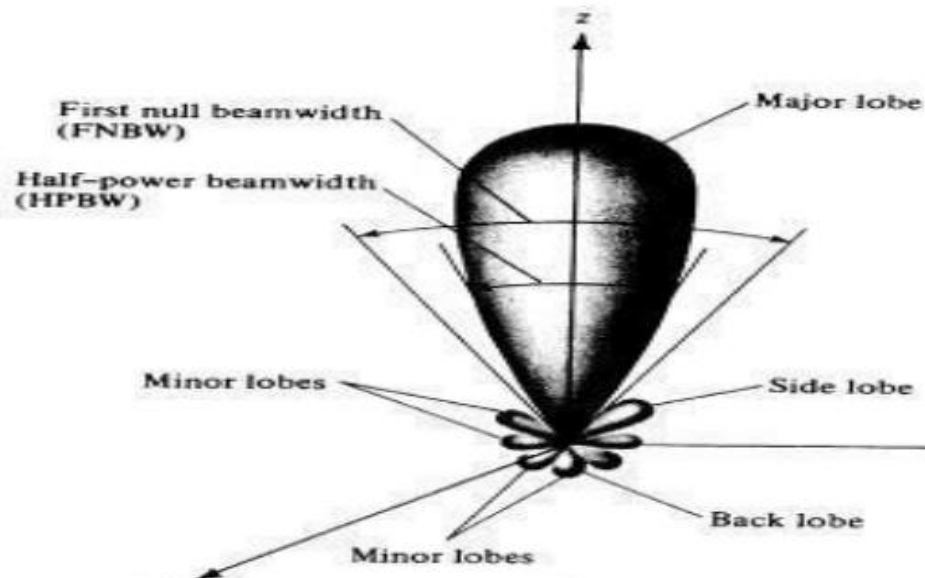
- According to their applications and technology available, antennas generally fall in one of two categories:
 1. Omnidirectional or only weakly directional antennas which receive or radiate more or less in all directions. These are employed when the relative position of the other station is unknown or arbitrary. They are also used at lower frequencies where a directional antenna would be too large, or simply to cut costs in applications where a directional antenna isn't required.
 2. Directional or *beam* antennas which are intended to preferentially radiate or receive in a particular direction or directional pattern.
-

Antenna Phase Functions:

- ➡ The phase function describes the 3-D radiation pattern of an antenna.
- ➡ Antenna patterns can produce very focused beams.
- ➡ Changing the way current flows in the antenna can change the **direction** as well!

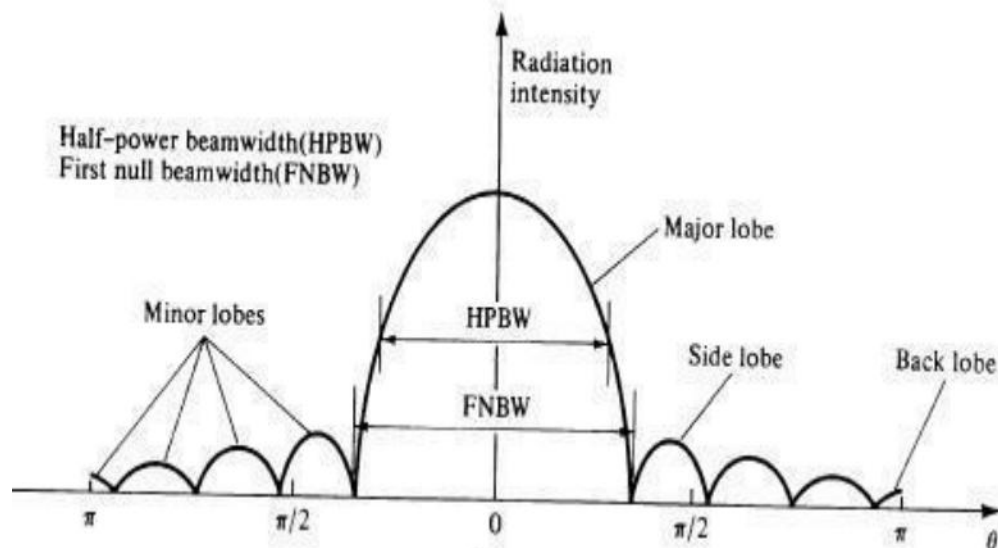


Pattern lobes



Pattern lobe is a portion of the radiation pattern with a local maximum

Lobes are classified as:
major, minor, side lobes, back lobes.



Antenna gain:



➡ The gain of an antenna is a directional measurement of its power relative to an isotropic radiator (such as a dipole). Like total power, Gain is typically given in units of decibels.

$$G_{\text{dB}} = 10 \log_{10}(P_{\text{out}}/P_{\text{iso}})$$

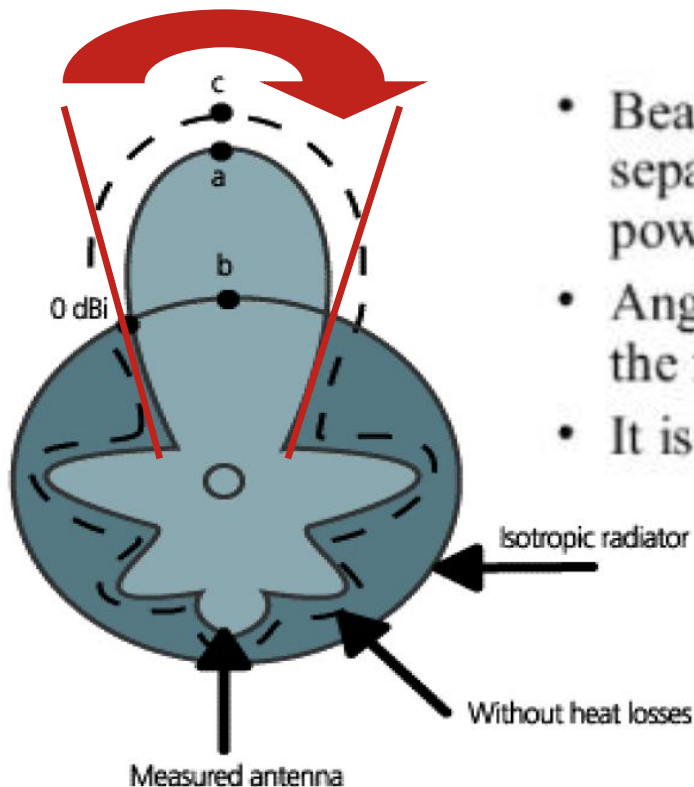
➡ A **positive** Gain means that a larger fraction of the radiated power is put in a given direction than in an isotropic system.

➡ A **negative** Gain means that a smaller fraction of the radiated power is put into a given direction than in an isotropic system.

Antenna Gain:



- ➡ A look back at the Friis function explains why Gain is important in communication. ($P_R \propto G^2$).
- ➡ The **beam width** (half power) is a measure of how directed the beam is.



- Beam-width of an antenna is defined as angular separation between the two half power points on power density radiation pattern OR
- Angular separation between two 3dB down points on the field strength of radiation pattern
- It is expressed in degrees



Effective Radiated Power

- The **effective radiated power** (ERP or EIRP) is the gain of an antenna (with respect to an isotropic radiator) multiplied by its input power.

$$\text{ERP} = \text{input power} \times \text{antenna gain}$$

- For example, a highly directional antenna with a gain of 7 has an input power of 1-kW. Its ERP is therefore 7 kW.
-

ISOTROPIC ANTENNA

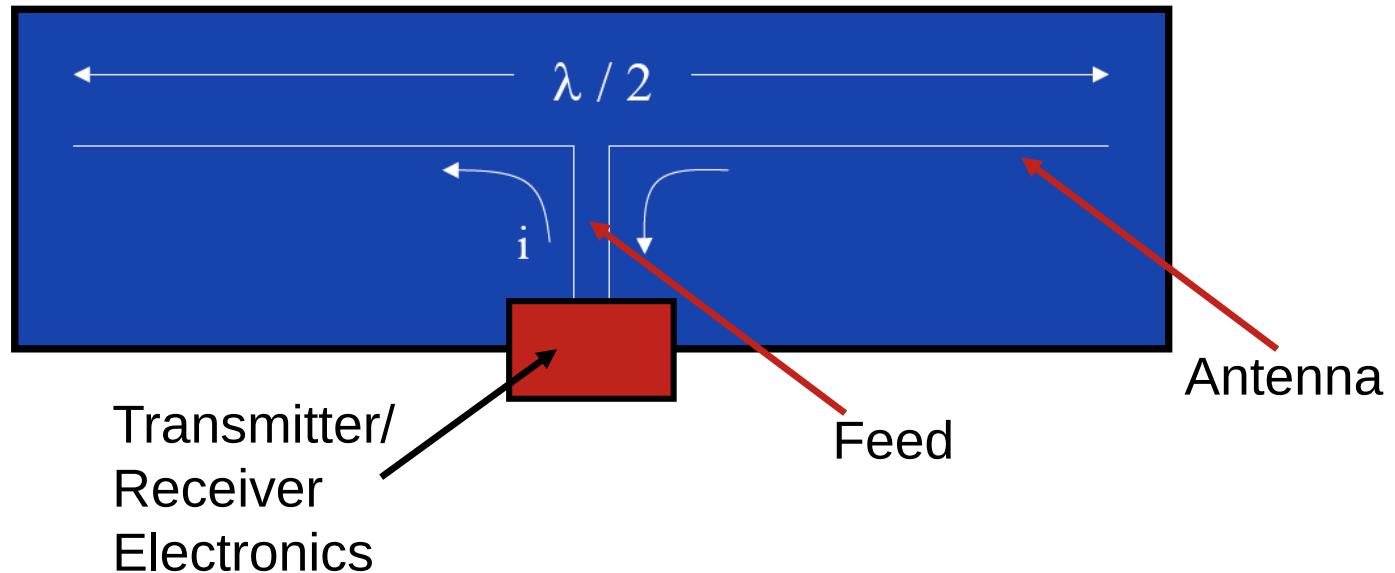


- *Isotropic antenna or isotropic radiator* is a hypothetical (not physically realizable) concept, used as a useful reference to describe real antennas.
- Isotropic antenna radiates equally in all directions.
 - Its radiation pattern is represented by a sphere whose center coincides with the location of the isotropic radiator.

The Dipole (Hertz) Antenna:



- ➡ The 1/2 wave Dipole Antenna is one of the simplest antenna designs.



- ➡ Dipole antennas radiate symmetrically away from the wires.



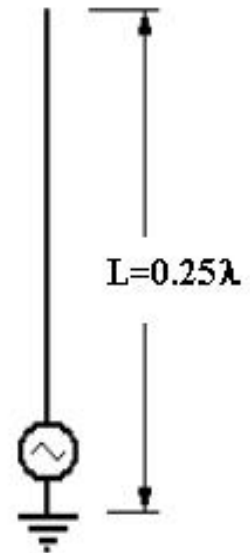
The Marconi (Vertical) Antenna:

➡ There are 3 issues with dipole antennas:

- 1) They can become VERY long. An AM dipole antenna would need to be **~400 feet long**.
- 2) To get good transmission along the ground requires mounting the antenna vertically.
- 3) Receiver antennas on cars must also be mounted vertically (to match **polarization**) and it is difficult to attach a 1/2 wave dipole.

➡ The Marconi Antenna is a solution:

- 1) The Marconi is essentially a 1/4 wave dipole.
- 2) It can be mounted at one spot.
- 3) It is half as long.



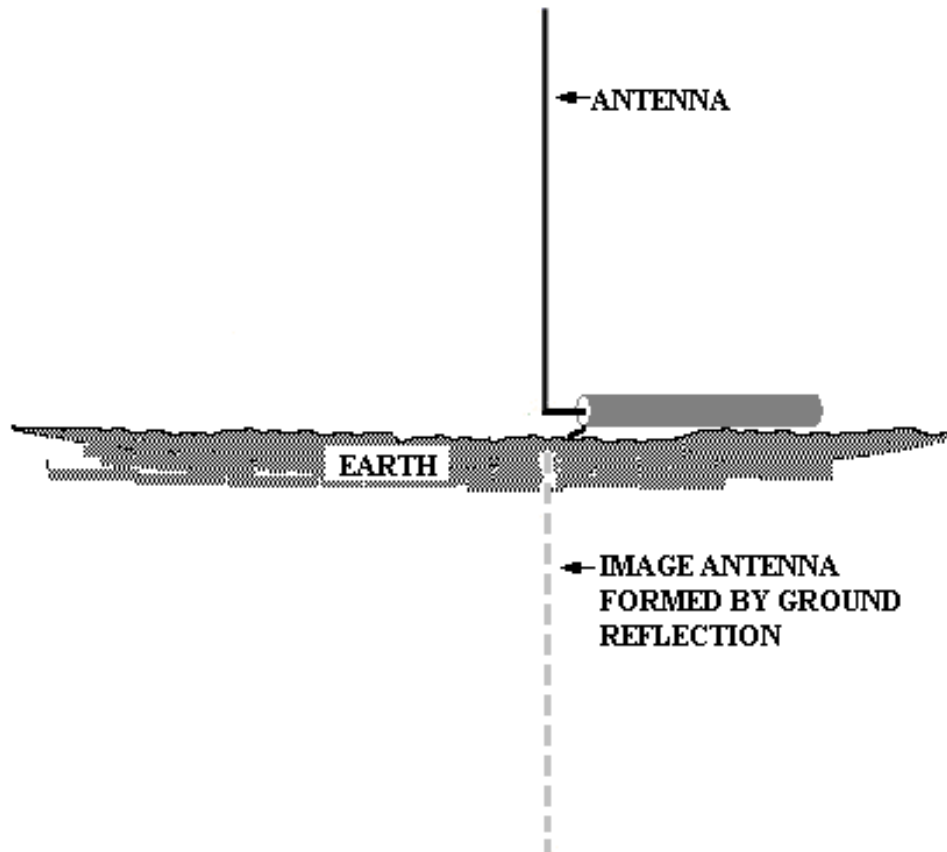
The Marconi Antenna:

innovate

achieve

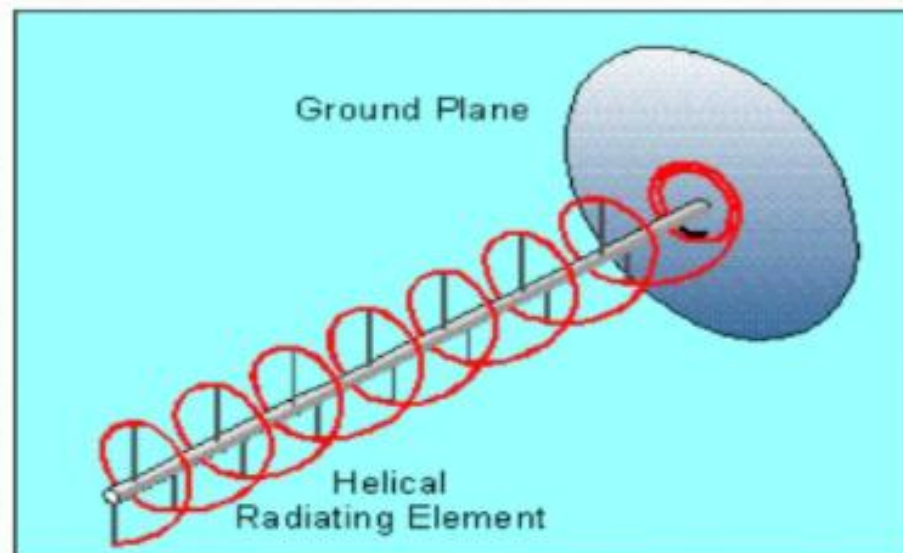
lead

- ➡ Marconi transmitting antennas take advantage of the reflective nature of the ground to RF:



HELIX ANTENNA

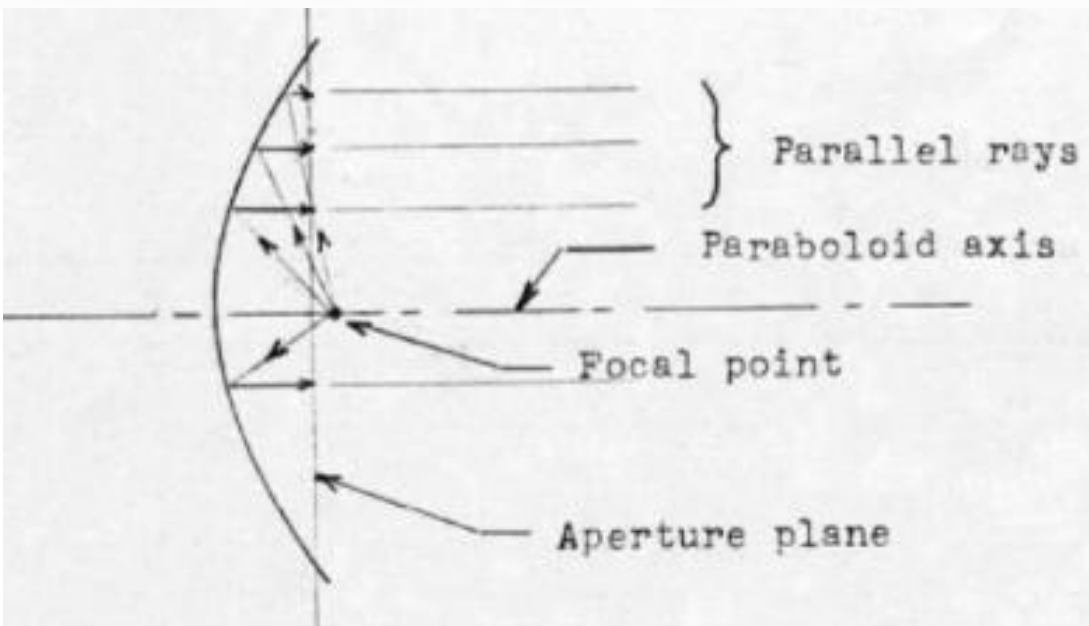
- A conducting wire in the form of screw thread fed by power source.
- The feed line is connected between the bottom of the helix and the ground plane.
- Helix antenna is Omni directional.
- Frequency range – VHF and UHF band.



helical antenna

PARABOLIC REFLECTOR ANTENNA

- This is used to convert spherical wave into plane wave
- The feed antenna is called primary antenna and reflector is secondary antenna.
- Frequency range - 3GHz-30GHz





ANTENNA ARRAYS

- Antenna arrays is group of antennas or antenna elements arranged to provide desired directional characteristics.
 - Generally any combination of elements can form an array.
 - However equal elements of regular geometry are usually used.
-

YAGI – UDA ANTENNA

- Yagi-Uda antenna consist of reflector, director and driven element.
- It is unidirectional antenna.
- Frequency range – 300MHz-3GHz



ANTENNA APPLICATIONS

They are used in systems such as

- Radio broadcasting
 - Broadcast television
 - Two-way radio
 - Communication receivers
 - Radar
 - Cell phones
 - Satellite communications.
-

ANTENNA CONSIDERATIONS

lead

- The space available for an antenna
- The proximity to neighbors
- The operating frequencies
- The output power
- Money